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Major: Quantitative Biomedicine

Project 2

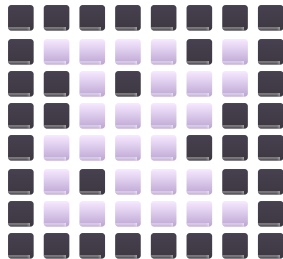
```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import random
from grid_generator import generate_ship
from baseline_strategy_2 import baseline_strategy_2
from developed_strategy import developed_strategy
from mobile_developed_strategy import mobile_developed_strategy
from Astar import a_star_output
from grid_generator import generate_ship, display_ship
```

Returning Number of moves from Project 1 Optimal strategy

```
In [2]: def print_report(title, res):
        print(f"\n {title} ")
        print(f"Total moves to localize (depth): {res['moves']}")
```

```
In [3]: D, seed = 8, 42
grid = generate_ship(D=D, seed=seed)
display_ship(grid)

# Optimal strategy: A* in belief space
opt_str = a_star_output(grid)
print_report("Optimal Strategy", opt_str)
```



Optimal Strategy
Total moves to localize (depth): 12

Answer to Question 4.4

Comparing the performance of Baseline strategy 2 and the Developed strategy (Third strategy)

```
In [4]: # Ship configuration
D = 15
grid = generate_ship(D, seed=85)
locator_pos = (2, 2)
roomba_pos = (10, 10)

# Range of detector sensitivities
```

```

alphas = [0.05, 0.1, 0.2, 0.3, 0.4, 0.5]

records = []

# Run experiments for each a value
for alpha in alphas:
    print(f'\nRunning experiments for  $\alpha$  = {alpha:.2f} ')

    # Setting seed for reproducibility
    random.seed(85)

    # Baseline Strategy 2
    res_base2 = baseline_strategy_2(grid, locator_pos, roomba_pos, alpha)

    # Resetting seed before running developed strategy
    random.seed(85)

    # Developed Strategy
    res_dev = developed_strategy(grid, locator_pos, roomba_pos, alpha)

    # Compute normalized utility (bounded between 0 and 1)
    # Higher means better (closer to 1)
    max_cost = res_dev['total_actions'] * 2 # simple normalization denominator
    if max_cost == 0:
        utility_norm = 0
    else:
        utility_norm = 1 - abs(res_dev['utility']) / max_cost
        utility_norm = round(max(0, min(1, utility_norm)), 3) # clamp to [0,1]

    # Record the two results
    records.append({
        'alpha': alpha,
        'strategy': 'Baseline 2',
        'moves': res_base2['moves'],
        'senses': res_base2['senses'],
        'total': res_base2['total_actions'],
        'utility': None, # baseline strategy has no utility
        'utility_norm': None
    })
    records.append({
        'alpha': alpha,
        'strategy': 'Developed',
        'moves': res_dev['moves'],
        'senses': res_dev['senses'],
        'total': res_dev['total_actions'],
        'utility': res_dev['utility'],
        'utility_norm': utility_norm
    })

# Convert to DataFrame and summarize
df = pd.DataFrame(records)
print('\nSummary Results')
print(df[['alpha', 'strategy', 'moves', 'senses', 'total', 'utility', 'utility_norm']])

```

Running experiments for $\alpha = 0.05$

Running experiments for $\alpha = 0.10$

Running experiments for $\alpha = 0.20$

Belief collapsed to (3, 4) (P=1.00) | Utility: -387.000

Running experiments for $\alpha = 0.30$

Running experiments for $\alpha = 0.40$

Belief collapsed to (2, 2) with P=2.84

Running experiments for $\alpha = 0.50$

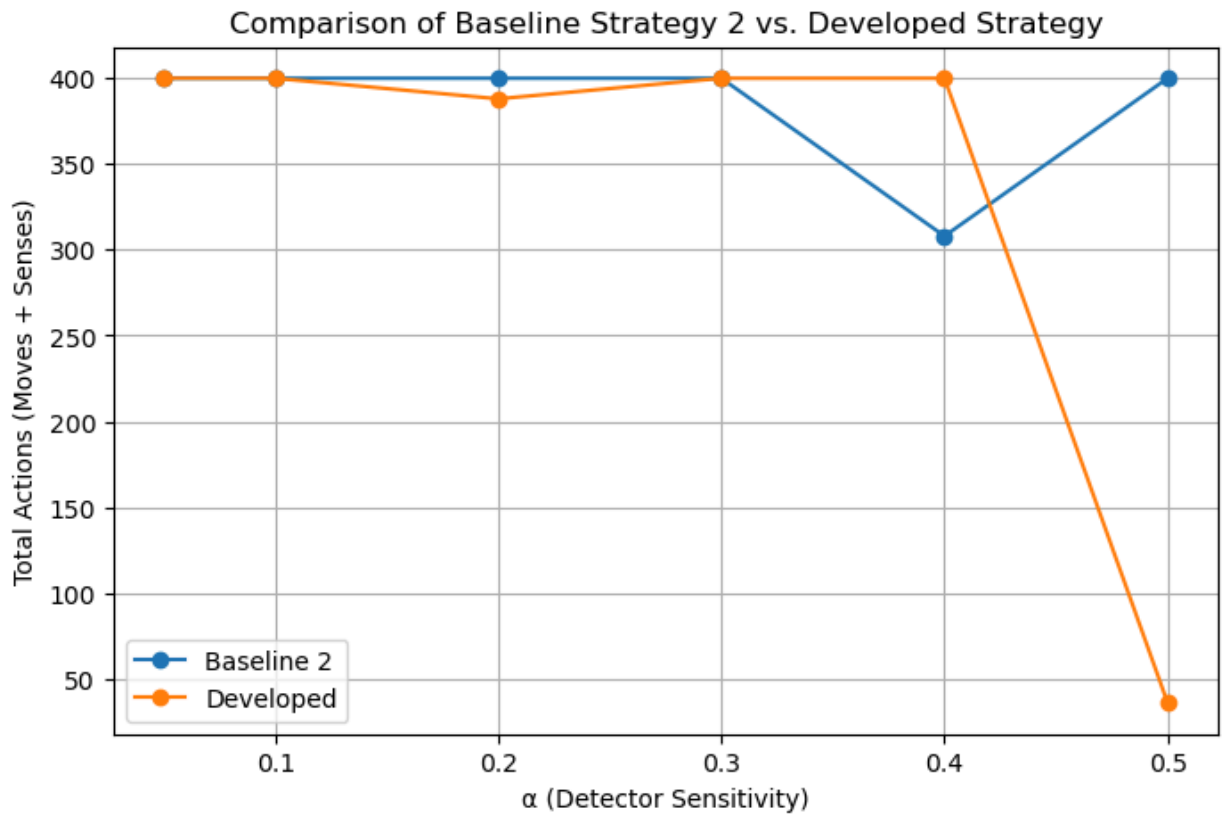
Belief collapsed to (6, 2) (P=1.00) | Utility: -35.000

Summary Results

	alpha	strategy	moves	senses	total	utility	utility_norm
0	0.05	Baseline 2	200	200	400	NaN	NaN
1	0.05	Developed	200	200	400	-406.495803	0.492
2	0.10	Baseline 2	200	200	400	NaN	NaN
3	0.10	Developed	200	200	400	-406.331529	0.492
4	0.20	Baseline 2	200	200	400	NaN	NaN
5	0.20	Developed	194	194	388	-387.000000	0.501
6	0.30	Baseline 2	200	200	400	NaN	NaN
7	0.30	Developed	200	200	400	-406.750772	0.492
8	0.40	Baseline 2	154	154	308	NaN	NaN
9	0.40	Developed	200	200	400	-406.020090	0.492
10	0.50	Baseline 2	200	200	400	NaN	NaN
11	0.50	Developed	18	18	36	-35.000000	0.514

```
In [5]: # Plot result
plt.figure(figsize=(8,5))
for strategy in ['Baseline 2', 'Developed']:
    df_sub = df[df['strategy'] == strategy]
    plt.plot(df_sub['alpha'], df_sub['total'], marker='o', label=strategy)

plt.xlabel('α (Detector Sensitivity)')
plt.ylabel('Total Actions (Moves + Senses)')
plt.title('Comparison of Baseline Strategy 2 vs. Developed Strategy')
plt.legend()
plt.grid(True)
plt.show()
```



Answer to Question 4.5

Comparing Baseline strategy 2, Third strategy, and Fourth strategy

```
In [6]: # Compare baseline strategy 2, Developed strategy, and Mobile strategy

D = 15
grid = generate_ship(D, seed=85)
locator_pos = (2, 2)
roomba_pos = (10, 10)

alphas = [0.05, 0.1, 0.2, 0.3, 0.4, 0.5]
records = []

for alpha in alphas:
    print(f"\nRunning experiments for  $\alpha$  = {alpha:.2f} ")

    # Reset random seed for reproducibility
    random.seed(85)

    # Baseline Strategy 2
    res_base2 = baseline_strategy_2(grid, locator_pos, roomba_pos, alpha)
    records.append({
        'alpha': alpha,
        'strategy': 'Baseline 2',
        'moves_R': res_base2['moves'],
        'moves_L': 0, # LocatorBot static
        'senses': res_base2['senses'],
        'total': res_base2['total_actions'],
        'utility': None
    })
```

```

}))

# Developed Strategy
random.seed(85)
res_dev = developed_strategy(grid, locator_pos, roomba_pos, alpha)
records.append({
    'alpha': alpha,
    'strategy': 'Developed',
    'moves_R': res_dev['moves'],
    'moves_L': 0, # LocatorBot static
    'senses': res_dev['senses'],
    'total': res_dev['total_actions'],
    'utility': res_dev['utility']
})

# Mobile Decision-Theoretic Strategy
random.seed(85)
res_mobile = mobile_developed_strategy(grid, locator_pos, roomba_pos, alpha)
records.append({
    'alpha': alpha,
    'strategy': 'Mobile Developed',
    'moves_R': res_mobile['moves_R'],
    'moves_L': res_mobile['moves_L'],
    'senses': res_mobile['senses'],
    'total': res_mobile['total_actions'],
    'utility': res_mobile['utility']
})

df = pd.DataFrame(records)
print("\nSummary Results")
print(df[['alpha', 'strategy', 'moves_R', 'moves_L', 'senses', 'total', 'utility']])

```

Running experiments for $\alpha = 0.05$

Running experiments for $\alpha = 0.10$

Running experiments for $\alpha = 0.20$

Belief collapsed to (3, 4) (P=1.00) | Utility: -387.000

Localized at (3, 5) after 26 R-moves, 74 L-moves, and 100 senses. | U=-199.292

Running experiments for $\alpha = 0.30$

Localized at (13, 6) after 25 R-moves, 25 L-moves, and 50 senses. | U=-100.049

Running experiments for $\alpha = 0.40$

Belief collapsed to (2, 2) with P=2.84

Localized at (11, 9) after 4 R-moves, 32 L-moves, and 36 senses. | U=-76.850

Running experiments for $\alpha = 0.50$

Belief collapsed to (6, 2) (P=1.00) | Utility: -35.000

Localized at (11, 8) after 8 R-moves, 37 L-moves, and 45 senses. | U=-92.398

Summary Results

	alpha	strategy	moves_R	moves_L	senses	total	utility
0	0.05	Baseline 2	200	0	200	400	NaN
1	0.05	Developed	200	0	200	400	-406.495803
2	0.05	Mobile Developed	156	44	200	400	-374.050371
3	0.10	Baseline 2	200	0	200	400	NaN
4	0.10	Developed	200	0	200	400	-406.331529
5	0.10	Mobile Developed	125	75	200	400	-376.481143
6	0.20	Baseline 2	200	0	200	400	NaN
7	0.20	Developed	194	0	194	388	-387.000000
8	0.20	Mobile Developed	26	74	100	200	-199.292365
9	0.30	Baseline 2	200	0	200	400	NaN
10	0.30	Developed	200	0	200	400	-406.750772
11	0.30	Mobile Developed	25	25	50	100	-100.049051
12	0.40	Baseline 2	154	0	154	308	NaN
13	0.40	Developed	200	0	200	400	-406.020090
14	0.40	Mobile Developed	4	32	36	72	-76.850311
15	0.50	Baseline 2	200	0	200	400	NaN
16	0.50	Developed	18	0	18	36	-35.000000
17	0.50	Mobile Developed	8	37	45	90	-92.397992

```
In [7]: # Plot results
try:
    import matplotlib.pyplot as plt
    plt.figure(figsize=(8,5))
    for strat in df['strategy'].unique():
        subset = df[df['strategy'] == strat]
        plt.plot(subset['alpha'], subset['total'], marker='o', label=strat)
    plt.xlabel('Detector Sensitivity ( $\alpha$ )')
    plt.ylabel('Total Actions')
    plt.title('Performance Comparison Across Strategies')
    plt.legend()
    plt.grid(True, linestyle='--', alpha=0.6)
    plt.show()
except Exception as e:
    print("Plot skipped (matplotlib not available):", e)
```

